Top 10 GO Biological Process Gene Sets Ranked by Potency of Perturbation (Sorted by BMD 25% Relative Deviation at the 5th Percentile)

DTTID: 108-020-004-000-5
Study Number:108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Ovary

Date: 09 Jun 2025
Time: 09:35:54 AM

Female

Category Name	Input Genes/Platform Genes in Gene Set	% Gene Set Coverage	BMD_{rd25} at the 5th Percentile (ppm)	$BMDL_{rd25}$ at 5th Percentile - $BMDU_{rd25}$ at 5th Percentile (ppm)	Genes with Changed Direction Up	Genes with Changed Direction Down
GO:0001667 ameboidal-type cell migration	5/55	9.1	223.1	90.9 - 432.8	4	1
GO:2001235 positive regulation of apoptotic signaling pathway	6/58	10.3	282.7	140.2 - 432.6	3	3
GO:0061041 regulation of wound healing	5/59	8.5	309.9	178.5 - 463.5	3	2
GO:0048639 positive regulation of developmental growth	5/50	10.0	332.8	176.6 - 485.1	3	2
GO:0030307 positive regulation of cell growth	3/47	6.4	335.6	203.0 - 460.4	2	1
GO:0015850 organic hydroxy compound transport	5/60	8.3	347.9	198.4 - 485.0	0	5
GO:0002526 acute inflammatory response	3/43	7.0	354.0	206.0 - 481.4	3	0
GO:2001242 regulation of intrinsic apoptotic signaling pathway	6/75	8.0	366.2	144.6 - NC	4	2
GO:0051385 response to mineralocorticoid	3/45	6.7	382.1	233.3 - 525.2	1	2
GO:1903034 regulation of response to wounding	5/69	7.3	383.0	243.8 - 527.7	3	2

rd25 = 25% relative deviation
NC = non-convergent

Active Genes:
GO:0001667: S100a9;Nrp1;Kdr;Cxcl12;Apoa1
GO:2001235: Vnn1;Tnfrsf12a;S100a9;Myc;Jak2;Ei24
GO:0061041: Tnfrsf12a;S100a9;Plg;F2r;Anxa5
GO:0048639: Tnfrsf12a;Nrp1;Mapk14;Cxcl12;Adrb2
GO:0030307: Tnfrsf12a;Nrp1;Cxcl12
GO:0015850: Slc51a;Slc10a2;Myc;Apoa1;Abcc4
GO:0002526: Vnn1;Hp;A2m
GO:2001242: Vnn1;S100a9;Myc;Herpud1;Ei24;Cxcl12
GO:0051385: Maob;Grin2b;Fosl1
GO:1903034: Tnfrsf12a;S100a9;Plg;F2r;Anxa5

Annotations
Official active gene symbols from the Rat Genome Database are shown in the "Active Genes" section above. Definitions of Gene Ontology terms were provided by the Gene Ontology Resource (http://geneontology.org/) and are included below.
GO process description version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

GO Biological Process Definitions:
GO:0001667 ameboidal-type cell migration: Cell migration that is accomplished by extension and retraction of a pseudopodium.
GO:2001235 positive regulation of apoptotic signaling pathway: Any process that activates or increases the frequency, rate or extent of apoptotic signaling pathway.
GO:0061041 regulation of wound healing: Any process that modulates the rate, frequency, or extent of the series of events that restore integrity to a damaged tissue, following an injury.
GO:0048639 positive regulation of developmental growth: Any process that activates, maintains or increases the rate of developmental growth.
GO:0030307 positive regulation of cell growth: Any process that activates or increases the frequency, rate, extent or direction of cell growth.
GO:0015850 organic hydroxy compound transport: The directed movement of an organic hydroxy compound (organic alcohol) into, out of or within a cell, or between cells, by means of some agent such as a transporter or pore. An organic hydroxy compound is an organic compound having at least one hydroxy group attached to a carbon atom.

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GO:0002526 acute inflammatory response: Inflammation which comprises a rapid, short-lived, relatively uniform response to acute injury or antigenic challenge and is characterized by accumulations of fluid, plasma proteins, and granulocytic leukocytes. An acute inflammatory response occurs within a matter of minutes or hours, and either resolves within a few days or becomes a chronic inflammatory response.

GO:2001242 regulation of intrinsic apoptotic signaling pathway: Any process that modulates the frequency, rate or extent of intrinsic apoptotic signaling pathway.

GO:0051385 response to mineralocorticoid: Any process that results in a change in state or activity of a cell or an organism (in terms of movement, secretion, enzyme production, gene expression, etc.) as a result of a mineralocorticoid stimulus. Mineralocorticoids are hormonal C21 corticosteroids synthesized from cholesterol and characterized by their similarity to aldosterone. Mineralocorticoids act primarily on water and electrolyte balance.

GO:1903034 regulation of response to wounding: Any process that modulates the frequency, rate or extent of response to wounding.

** END OF REPORT **